DEVELOPMENT ACTIVITY 3 — PARALLEL MODULE DEVELOPMENT & INTEGRATION

FairFarm — Farmer Marketplace with Price Transparency | Secure Software Development

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Submission Overview & Project Information

Project Name: FairFarm — Farmer Marketplace with Price Transparency

Group Members: Abdul-Wahab (2026-SSD-01), Ahmad-Fareed (2026-SSD-02)

Public URL: https://fair-farm-marketplace-p81iwnhzy-ogbrutal2825-8857s-projects.vercel.app/

Module 1: Submit Crop Listing (SubmitListing.jsx) — Developed by Abdul-Wahab

Module 2: Browse & Search Crop Listings (BrowseListings.jsx) — Developed by Ahmad-Fareed

Part 1 — Selected Complementary Modules

Module 1 (Submit Crop Listing): Enables farmers to list agricultural produce with complete quality, quantity, price, and district metadata.

Module 2 (Browse & Search Listings): Enables buyers and officers to explore listings using real-time search, quality/status filters, sorting, and modal details.

Part 2 — Member 1: Develop Module 1 (Abdul-Wahab)

Responsible Member: Abdul-Wahab

Primary Function: Produce submission form with live quality & price input fields.

Validation Rules: Mandatory fields, positive numeric checks (> 0), min 2 chars for variety.

JS Interactions: Real-time validation feedback, stateful session listing list, toast notification.

Part 3 — Member 2: Develop Module 2 (Ahmad-Fareed)

Responsible Member: Ahmad-Fareed

Primary Function: Interactive produce catalog with multi-filter and inspection modal.

Validation / Search Rules: Multi-field search, status/quality filter chips, empty result recovery.

JS Interactions: Dynamic array filtering/sorting, filter chip removal, animated inspection modal.

Part 4 — Integration Checklist

• Module 1 accessible at /submit-listing for authenticated Farmers.

• Module 2 accessible at /browse-listings for all users.

• Integrated into main Navbar and App router with unified brand theme.

Part 5 — Testing Matrix & Issues Discovered

Test Case	Expected Result	Actual Result
Submit valid listing	Listing added, toast shown	Pass - Appended to sidebar list
Leave required empty	Form blocked, red error shown	Pass - Error messages displayed
Search & filter grid	Grid updates reactively	Pass - Cards filtered instantly

Module 1 Issue Discovered: Inputs remained in form after submission. *Fix: Added automatic form state reset and success toast.*

Module 2 Issue Discovered: Empty search query broke layout. *Fix: Added dedicated empty state component with reset button.*

Part 6 — Security-Aware Development

Module 1 Misuse & Response: Negative pricing / XSS injection -> Sanitization + positive numeric validation (> 0) + Pydantic backend check.

Module 2 Misuse & Response: Bidding on closed/sold items -> Bidding state lock disabling action + server transaction verification.

JavaScript Interactions Summary

1. Real-time form validation with inline errors and session listing state push.
2. Multi-criteria search, active filter chips, price sorting, and animated modal dialog.